Adam Cool

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Education

McMaster University Hamilton, ON
Bachelor of Engineering, Computer Co-op *September 2015 – April 2021*

- Programming projects and assignments in Python, Java, C, Assembly, and Verilog.
- Completed challenging projects with other students and clients: one-handed apple slicer, FPGA programmed image encoder/decoder, as well as a machine learning backed data analysis project for RBC.
- Studies focused on parallel processing design, hardware architecture, algorithm design and analysis, data structures, and data encoding/decoding.

Experience

Family Businesses May 2021 - Current
Contract Worker *Remote*

- Managed various computer related projects: IT systems, script writing, website management, VR flight simulator.
- Provided support in on-site environments: construction sites, shipyards, and electronics repair.

Advanced Micro Devices (AMD) May 2018 - August 2019
Software Developer Co-op *Markham, Ontario*

- Completed 2 projects over the work term: Machine Learning proof of concept and a Memory Access Driver.
- Brainstorm, design, and prototype a proof-of-concept using Python Tensorflow to show that AMD GPUs were capable of GPU accelerated learning.
- Designed a remote-control car that could self-steer within roadways and identify signs and obstacles.
- Agile work environment using JIRA to manage tasks and Git version control on the code base.
- Used Confluence for finding and sharing information between members of the team.
- Programmed and debugged driver for managing memory between multiple embedded processing units on an SOC.
- Worked closely with a team of senior developers to build automated tests and integrate the driver into a larger program built on top of a Unix distro.

Hybrid Enterprises, Lockheed Martin April - August 2016
Flight Simulation Model Designer *Edmonton, Alberta (Remote)*

- Designed and built portable flight simulator unit for Lockheed Martin Hybrid Airship.
- Assembled hardware for the system, and ported an existing hybrid airship flight model to new simulator software.
- Created flight simulator booth for use at conventions and showings of the Hybrid Airship.
- Represented project at 2016 Abbotsford International Airshow operating the Hybrid Airship flight simulator demo.

Student Tutoring 2012 - Current
Tutor for Math and Science

- Tutored students in both Highschool and University with both Math and Science.
- Continued to provide tutoring to my younger family members currently going through Highschool.

Technical Skills

Operating Systems: Windows, MacOS, Unix, Linux Distributions (Red Hat, Ubuntu, Mint)
Languages: Verilog, C/C++, Java, Python, SQL, JavaScript, HTML/CSS/PHP, Racket
Developer Tools: Git, JIRA, Jenkins, Docker, PyCharm, VS Code, Visual Studio, Codeblocks, Eclipse, Unity, Unreal Engine, Godot
Libraries: Numpy, Pandas, Matplotlib, Tensorflow, Pytorch, OpenGL
Multimedia: Microsoft Office, Photoshop, Excel, AutoCAD, Blender, 3DS Max
Hands-on: Electrical component soldering, Basic Woodworking, 3D printing
Course Work: State-based design and programming, High level calculus and algebra, Analysis of fields in 3-dimensional space, Statistical analysis and probabilities, Classical and quantum physics, Chemistry and Materials, Radio signal analysis and processing, Engineering design, Operating system Design, Parallel Processing, Circuit and gate level analysis and design